

Anti-angiogenic activities of *Cnidium officinale* Makino and *Tabanus bovinus*

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Abstract

This study investigated the anti-angiogenic activities of *Cnidium officinale* Makino and *Tabanus bovinus* by using cultured glomerular capillary endothelial cells (GECs), chorioallantoic membrane (CAM) and rat cornea. Treatment of GECs with several concentrations (5–50 µg/ml) of *C. officinale* Makino and *T. bovinus* extracts for 24 h inhibited angiotensin II (10^{-8} M)-induced increases of [³H]thymidine uptake and cell numbers in a concentration-dependent manner. The extent of inhibitory rate of [³H]thymidine incorporation by *C. officinale* Makino and *T. bovinus* at 50 µg/ml was a similar to that by 10^{-5} M of retinoic acid. Herbal extracts also conspicuously inhibited the neovascularization. In contrast to the normal branching of vascular vessels, blood vessel patterns in CAMs treated with extracts (50 µg per egg) of *C. officinale* Makino and *T. bovinus* were ran parallel to each other without much branching. Moreover, oral administration of herbal extracts (20 mg/kg per day) for 4 weeks significantly inhibited the rat corneal neovascularization induced by suture, and the length of blood vessels in herbal medicine-treated rat cornea was conspicuously lower than that in control animals. A similar inhibitory effect to these was also observed in the rat cornea treated with thalidomide (200 mg/kg per day). These findings indicate that the anti-angiogenic properties of *C. officinale* Makino and *T. bovinus* may be one of the pharmacological mechanisms underlying the anti-tumor and anti-metastatic activities of herbal extracts tested in this study. © 2002 Published by Elsevier Science Ireland Ltd.

Keywords: *Cnidium officinale* Makino; *Tabanus bovinus*; Angiogenesis

1. Introduction

The formation of new blood vessels out of pre-existing capillaries, angiogenesis, is a sequence of events that is of key importance in a broad array of physiologic and pathologic processes. In several diseases, excessive angiogenesis is part of the pathology, and these diseases

include atherosclerosis, rheumatoid arthritis, Crohn's disease, diabetic retinopathy and solid tumors and its metastasis (Griffioen and Molema, 2000). In cancer, growth of solid tumor cells subsequently leads to initiation of angiogenesis (Holash et al., 1999), and circulating endothelial progenitor cells can form an additional source for postnatal vasculogenesis in tumor growth (Asahara et al., 1999). Therefore, angiogenesis therapy is considered a promising approach, supposedly leading to the desperately needed breakthrough in cancer therapy and other proangiogenic diseases.

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Cnidium officinale Makino and *Tabanus bovinus* are known to have a regulatory property for blood circulation and inflammatory diseases (Tahara et al., 1999; Wang et al., 1989). These natural products are one of the major herbs used in oriental medicine to treat pain, inflammation and menstrual disturbance; these actions are probably produced by transient vasodilation. Higashi (1996) suggested that *C. officinale* Makino may induce the enhancement or activation of blood congestion and inflammation and may cause itching based on the promotion of blood circulation in inflammatory diseases. In addition, Haranaka et al. (1985) and Onishi et al. (1998) also reported the anti-tumor and anti-metastatic activities of *C. officinale* Makino in ddY mice and hepatic carcinoma cells, respectively. Recently, a potent anti-cancer activity of *T. bovinus* has been demonstrated by using HeLa cells originated from human cervix and uterine cancer (Ahn et al., 2000).

Regarding these suggestions related to the pathological role of angiogenesis and the anti-tumor activities of *C. officinale* Makino and *T. bovinus*, we hypothesize that the anti-tumor and anti-metastatic properties of *C. officinale* Makino and *T. bovinus* might be mediated by anti-angiogenic mechanism. To examine this hypothesis, the present study investigated the anti-angiogenic activities of *C. officinale* Makino and *T. bovinus* by using cultured renal glomerular capillary endothelial cells (GECs), chick embryo chorioallantoic membrane (CAM) and rat cornea.

2. Materials and methods

2.1. Materials

Fertilized chick eggs were obtained from Youngsang Farm (Jeonbuk, South Korea). Fat emulsion (10%) was from Green Cross Pharm. Co. (Seoul, South Korea). Thermanox coverslips and 10-0 nylon were purchased from Nunc Inc. (Naperville, IL) and Davis+Geck (Wayne, NJ), respectively. Retinoic acid and collagenase (type IV) were supplied by Sigma Chemical Co. (St. Louis, MO). All other chemicals were of the highest grade from commercial sources.

2.2. Preparation of herbal extracts

C. officinale Makino and *T. bovinus* were purchased from the Korea Medicinal Herb Association and authenticated by Dr Won Hong Woo, College of Oriental Medicine, Wonkwang University. Voucher specimens (No. 2000-17 and 2000-39) were deposited at the Herbarium of the College of Natural Sciences, Wonkwang University. Extracts of the *C. officinale* Makino and *T. bovinus* were prepared by decocting the herbs with ten times (v/w) of distilled water for 1.5 h.

The residue was boiled for additional 1 h. Extracts were combined together and filtered through a 0.45 μm filter, lyophilized by freezing drier (Labconco, Preezone), and kept at 4 °C until use. The yields of extract were 12.5–15% of dried ingredient weight.

2.3. Cell culture

GECs were cultured by standard method as described previously (Laulajainen et al., 1993). Briefly, glomeruli from male Sprague–Dawley rats (5-week old) were isolated under sterile conditions by a sieving technique, and the resulting preparation contained greater than 95% glomeruli as judged by light microscopy. Glomeruli were treated with 0.1% collagenase for 20 min at 37 °C and washed twice in Dulbecco's modified Eagle's medium (DMEM, Gibco BRL) with 10% heat-inactivated FCS, 2 mM supplemental glutamine, 5 $\mu\text{g}/\text{ml}$ insulin, 100 U/ml penicillin and 100 $\mu\text{g}/\text{ml}$ streptomycin. Fifty microliters of the cell suspension were then plated into each well of a 96-well cell culture plate (Nunc) previously coated with fibronectin. The microtiter plates were screened by phase-contrast microscopy for the presence of cells with a cobblestone appearance, which is one of characteristic feature of cultured GECs (Baller-mann, 1989; Laulajainen et al., 1993). Cells fulfilling this criterion were subsequently cloned twice with limiting dilution, and a homogenous cell line was obtained. For all experiments, cells between passage 8 and 12 were used.

To identify biochemical characteristics of GECs, angiotensin I-converting enzyme (ACE) activities were measured in cell lysates with a commercial assay system (Sigma). This assay is a spectrophotometric method utilizing the hydrolysis of the synthetic peptides FAPGG by ACE. Human serum of known ACE activity served as a standard. For these experiments, primary culture of glomerular mesangial cells (passage 10–12) isolated from Sprague–Dawley rat was used as an additional control. ACE activities of GECs and mesangial cells were 0.12 ± 0.05 pmol/ 10^6 cells per min and 0.53 ± 0.07 pmol/ 10^6 cells per min, respectively.

2.4. Measurement of cell proliferation

Inhibitory effects of *C. officinale* Makino and *T. bovinus* on GECs proliferation were assayed by measuring [^3H]thymidine uptake and cell numbers. A total of 1×10^4 GECs was transferred to each well of a 24-well culture plate. After 24 h, cell growth was arrested for 48 h in DMEM containing 0.2% FCS. During serum deprivation, GECs were exposed for 16 h to fresh medium without serum and then incubated for 24 h with 10^{-8} M of angiotensin II (ANG II) in the absence or presence of several different concentration of herbal extracts (5–50 $\mu\text{g}/\text{ml}$). For determination of [^3H]thymi-

dine incorporation, GECs were pulsed with 1 $\mu\text{Ci/ml}$ [*methyl*- ^3H]thymidine (specific activity, 5 mCi/mmol; Amersham, Braunschweig) during the last 6 h of culture. To identify the cellular mechanisms involved in the inhibitory effects of *C. officinale* Makino and *T. bovinus* extracts on ANG II-induced GECs proliferation, this study employed losartan (10^{-6} M) and retinoic acid (10^{-5} M) as a drug of positive control. At the end of the incubation period, GECs were washed twice with phosphate-buffered saline (PBS, pH 7.4), incubated with 5% trichloroacetic acid for 30 min at 4 °C, washed with ethanol, and dissolved in 0.1 M NaOH (Meivar-Levy et al., 1997). Radioactivity was measured in a liquid scintillation counter (Beckman, Munich) using Ultima gold scintillation fluid (Packard):water (8:1, v/v). Cell numbers were also determined at the end of incubation period. Monolayers of GECs were washed twice in PBS, and cells were trypsinized and transferred into 10 ml of isoton solution (Coulter, Luton) for counting in a cell counter (Coulter).

2.5. Chorioallantoic membrane (CAM) assay

Anti-angiogenic activities of herbal extracts were measured by CAM assay with slight modification of the methods previously described (Oikawa et al., 1991). Briefly, the fertilized chicken eggs were kept in a humidified egg incubator at 37 °C. After 4 days incubation, about 3 ml of albumin was aspirated from eggs with an 18-gauge hypodermic needles through a small hole drilled at the narrow end of the eggs, allowing the small CAM and yolk sac to drop away from the shell membrane. On day 5, the shell covering the air sac was punched out and removed by forceps, and the shell membrane on the floor of the air sac was peeled away. Embryos with chorioallantois of 3–5 mm in diameter were employed for the assay of antiangiogenic activity. Five microliters of herbal extracts (50 μg per egg) or retinoic acid (10 μg per egg) as a positive control were applied to sterile Thermanox 15 mm disks and allowed dry under laminar flow conditions. The loaded disks were inverted and applied to the CAM surface of 5.5-day-old embryos through the window. The air sac ends of the embryos with shells were covered with scotch tape. Two days later, an appropriate volume of a 10% fat emulsion was injected by using a 33-gauge needle into the 7.5-day-embryo chorioallantois so that the vascular network of CAM stood out against the white background of lipid. At least 20 eggs were used for each condition.

2.6. Cornea neovascularization

To clarify effects of *C. officinale* Makino and *T. bovinus* on neovascularization, this study used rat corneal model induced by suture method, which has

been previously described in detail (Park et al., 1998). All animals were maintained under 12-h light–dark cycles with temperature of 23 ± 2 °C and relative humidity of 55–60% and allowed free access to a commercial diet (Samyang, Seoul, South Korea) and drinking water. Briefly, after male Sprague–Dawley rats (200–250 g) were anesthetized with pentobarbital (Nembutal, Abbott, Chicago), corneal neovascularization was induced by suturing (one stitch) the cornea with 10-0 nylon, and teramycin was applied on the cornea for 1 week to prevent corneal inflammation. To study the anti-angiogenic activities of *C. officinale* Makino and *T. bovinus*, herbal extracts were orally administered at 20 mg/kg per day for 4 weeks. Control rats orally received an equal volume of water instead of the herbal extracts, and rats administered orally with thalidomide (200 mg/kg) were also employed as a positive control. Corneal neovascularization and vessel lengths were photographed and measured at various time points (3 days–4 weeks) after induction of neovascularization.

2.7. Statistical analysis

Values obtained in this study are expressed as mean $\pm$ S.E. Statistical comparison was made by one-factor analysis of variance and Student–Newman–Keuls procedure, and statistical difference was considered to be significant at $P < 0.05$.

3. Results

3.1. Inhibition of cell proliferation

Since it is generally accepted that ANG II is a potential growth factor for GECs (Wolf et al., 1996), the inhibitory effects of *C. officinale* Makino and *T. bovinus* extracts on ANG II-induced GECs proliferation were examined. ANG II (10^{-8} M) significantly increased the [^3H]thymidine uptake into quiescent GECs (1.6 ± 0.09 -fold, $P < 0.05$), and this increase was attenuated by the treatment of GECs with *C. officinale* Makino and *T. bovinus* (5–50 $\mu\text{g/ml}$) in a concentration-dependent manner (Fig. 1A). *C. officinale* Makino and *T. bovinus* extracts at 30 $\mu\text{g/ml}$ diminished the ANG II-induced increase of [^3H]thymidine uptake to $56.3 \pm 6.72\%$ ($P < 0.01$) and $65.2 \pm 6.36\%$ ($P < 0.05$), respectively. A similar phenomenon of inhibitory properties was also observed in the changes of cell numbers after the treatment of GECs with herbal extracts (data not shown).

Furthermore, this study examined whether anti-angiogenic mechanism might be involved in the inhibitory effects of *C. officinale* Makino and *T. bovinus* on ANG II-induced GEC proliferation by using losartan, a specific AT₁ receptor blocker (Wolf et al., 1996), and

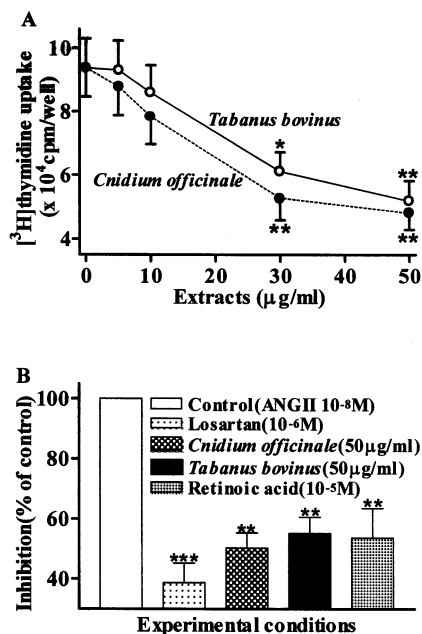


Fig. 1. Concentration–response (A) and inhibitory effects (B) of *C. officinale* Makino and *T. bovinus* on ANG II-induced increases of $[^3\text{H}]$ thymidine uptake in cultured GECs. Experiments were independently repeated six times with duplicate measurements for each sample, and results are expressed as mean $\pm$ S.E. *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$ vs. control cells treated with ANG II alone.

retinoic acid, a potent anti-angiogenic agent (Belloni et al., 2000). As shown in Fig. 1B, ANG II-stimulated incorporation of $[^3\text{H}]$ thymidine was dramatically reduced by treatment of GECs with losartan (10^{-6}M , $P < 0.001$), and also significantly inhibited by treatment of GECs with retinoic acid (10^{-5}M , $P < 0.01$). The extent of inhibitory rate of $[^3\text{H}]$ thymidine incorporation provided by *C. officinale* Makino and *T. bovinus* at 50 $\mu\text{g/ml}$ was essentially a similar to the extent of inhibitory rate of it by 10^{-5}M of retinoic acid.

3.2. Inhibition of chorioallantoic membrane angiogenesis

CAM assay was employed for examining the inhibitory properties of *C. officinale* Makino and *T. bovinus* extracts on vascular development, and retinoic acid was used as a positive control for this experiment. The area of the CAM below disks without treatment did not show changes in vascular density, and a normal branching pattern of blood vessels was present (Fig. 2A), indicating that the disk weight did not affect their growth. After 48 h of treatment, branching pattern of blood vessels below disks containing retinoic acid (10 μg per egg) was dramatically decreased (Fig. 2B). A similar angiogenic inhibition was observed in the CAM treated with 50 μg per egg of *C. officinale* Makino and *T. bovinus* (Fig. 2C) and *T. bovinus* (Fig. 2D) extracts. Branching pattern of blood vessels between *C. officinale*

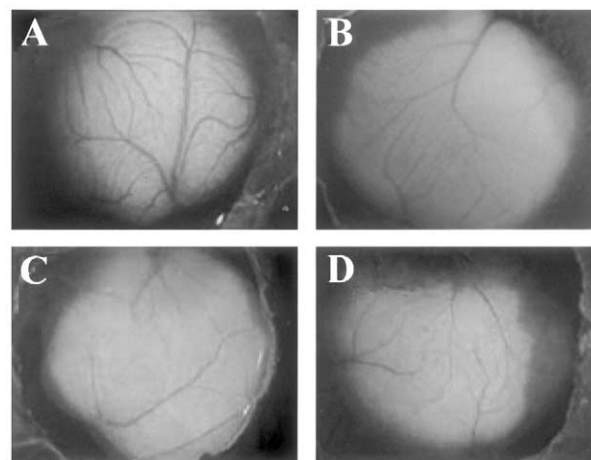


Fig. 2. Effects of *C. officinale* Makino and *T. bovinus* on embryonic angiogenesis in CAM 2 days after sample implantation. Control CAMs treated with empty coverslips show no disturbance of angiogenesis (A). CAMs implanted with coverslips loaded with retinoic acid (10 $\mu\text{g/ml}$, B), *C. officinale* Makino (50 $\mu\text{g/ml}$, C) and *T. bovinus* (50 $\mu\text{g/ml}$, D).

Makino- and *T. bovinus*-treated CAM was not significantly different.

3.3. Inhibition of cornea neovascularization

This study continued to examine the anti-angiogenic activity of *C. officinale* Makino and *T. bovinus* on neovascularization in rat corneal model produced by suture. In the cornea of rats untreated with herbal extracts, numerous vessels were usually seen invading the cornea by day 2 or 3 after suture, and reaching the lesion area within 5–6 days (Fig. 3A). In contrast to this, oral administration of *C. officinale* Makino and *T. bovinus* extracts (20 mg/kg per day) for 4 weeks induced a marked reduction of vascular formation in cornea (Fig. 3B and C).

When the dense vascular networks extending from the corneoscleral limbus were measured at 1–4 weeks, the length of blood vessels in the neovascularized cornea of *C. officinale* Makino and *T. bovinus*-administered rats markedly reduced compared with that in control animals ($P < 0.001$, Fig. 4). By using thalidomide, a potent angiogenic inhibitor (D'Amato et al., 1994), potency of inhibitory effects of *C. officinale* Makino and *T. bovinus* on corneal neovascularization was estimated. Administration of thalidomide (200 mg/kg per day) significantly inhibited the suture-induced corneal neovascularization, and its inhibitory rate was not significantly different from that of *C. officinale* Makino and *T. bovinus* at concentrations used here. These results are consistent with the previous results shown in Fig. 2, suggesting that extracts of *C. officinale* Makino and *T. bovinus* might have properties to inhibit the formation of new vascular vessels.

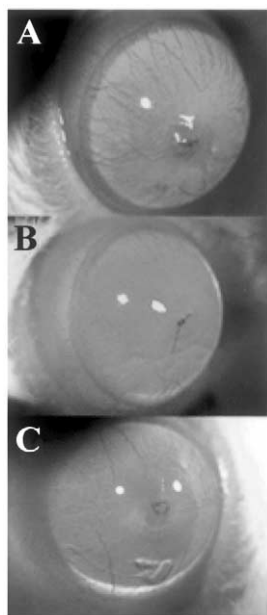


Fig. 3. Inhibitory effects of *C. officinale* Makino and *T. bovinus* on the neovascularization of rat cornea. Animals were orally treated with herbal extracts (20 mg/kg body weight per day) for 4 weeks, and slitlamp photographs of cornea were taken. Figures are showing an intensive corneal neovascular response by suture (A) and a marked inhibition of neovascularization by *C. officinale* Makino (B) and *T. bovinus* (C).

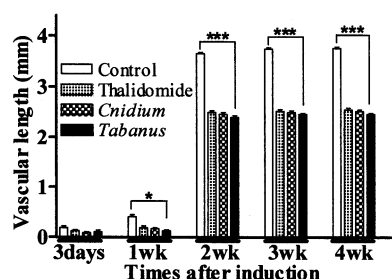


Fig. 4. Effects of *C. officinale* Makino and *T. bovinus* on vascular length in rat cornea. Animals were orally treated with each herbal extracts (20 mg/kg per day) or thalidomide (200 mg/kg per day) for 4 weeks. The length of vascular extension from limbus was measured with triplicate for each cornea at various times. Results are expressed as mean $\pm$ S.E. of ten to 12 rats. **, $P < 0.01$; ***, $P < 0.001$ vs. control for each time.

4. Discussion

The present study showed that extracts of *C. officinale* Makino and *T. bovinus* inhibited the proliferation of GECs and angiogenesis of chick CAM. Furthermore, oral administration of *C. officinale* Makino and *T. bovinus* also markedly inhibited the neovascular formation in rat cornea. These findings may support the previous reports (Haranaka et al., 1985; Onishi et al., 1998), suggesting that anti-metastatic and anti-tumor activities of *C. officinale* Makino are based on the

regulation of reticuloendothelial system. Therefore, results obtained in this study might indicate that one mechanism underlying anti-metastatic and anti-tumor activities of *C. officinale* Makino and *T. bovinus* might be mediated, in part, by anti-angiogenic activities.

Angiogenesis is characterized by increased proliferation of microvessel endothelial cells, and the uncontrolled endothelial cell proliferation takes place during tumor neovascularization and in several angioproliferative diseases, including diabetic retinopathy, arthritis and vascular tumors. A potential model for the sequence of events involved in angiogenesis is one which angiogenic factors stimulate the proliferation of capillary endothelial cells to protrude proteolytic enzymes such as collagenase and plasminogen activators. In cultured capillary endothelial cells originated from renal glomeruli, ANG II stimulates endothelial proliferation through production of many cytokines and growth factors (Wolf et al., 1996). In agreement with this suggestion, ANG II (10^{-8} M) stimulated the GECs proliferation which was shown in increases of [3 H]thymidine incorporation and cell numbers, and this stimulation was inhibited by treatment of GECs with extracts of *C. officinale* Makino and *T. bovinus*. Since concentrations of herbal extracts used in this study did not change the viability of endothelial cells (data not shown), it is difficult to conclude that inhibitory effects of *C. officinale* Makino and *T. bovinus* on ANG II-stimulated endothelial proliferation may be related to the cytotoxicity caused by herbal medicines. Therefore, although the present study could not demonstrate a detailed pharmacological mechanism involved in the inhibitory effects of *C. officinale* Makino and *T. bovinus* on ANG II-induced GECs proliferation, the results obtained in this study suggest that *C. officinale* Makino and *T. bovinus* may have an anti-proliferative properties of endothelial cells. Furthermore, this study also examined the anti-proliferative activities of *C. officinale* Makino and *T. bovinus* on growth of four different types of cancer cells such as SNU 638 (gastric carcinoma cells), SNU 449 (hepatic carcinoma cells), ECV 304 (endothelial carcinoma cells) and HL-60 (promyelocytic leukemia cells). Among these cancer cells, tumor growth rate was inhibited more effectively in endothelial carcinoma cells, ECV 304, by *C. officinale* Makino and *T. bovinus* extracts (data not shown). A significant inhibition could be detected as low as 100 μ g/ml of both herbal extracts. It has been well known that retinoic acid is an effective chemopreventive agent against squamous cell carcinoma and can inhibit angiogenesis induced by tumor cells (Lingen et al., 1996). Therefore, this study used retinoic acid as a positive drug to estimate inhibitory properties of *C. officinale* Makino and *T. bovinus* on proliferation of capillary endothelial cells (Fig. 1B). The extent of inhibitory rate of [3 H]thymidine incorporation provided

by *C. officinale* Makino and *T. bovinus* at 50 µg/ml was essentially a similar to the extent of inhibitory rate of it by 10^{-5} M of retinoic acid. Based on these, it is conceivable that inhibitory effects of *C. officinale* Makino and *T. bovinus* on proliferation of capillary endothelial and cancer cells may be related to the anti-angiogenic activities.

The availability of chemical agents that could prevent the continued spread of vascularization would potentially have a broad applicability as a novel therapy for tumor growth inhibition. Onishi et al. (1998) suggested that *C. officinale* Makino showed a significant inhibition of liver metastasis by colon 26-L5 carcinoma cells as well as lung metastasis by B16-BL6 melanoma cells in vivo. It is also suggested that anti-tumor activity of *C. officinale* Makino is based on the stimulation of reticuloendothelial system and is closely related to the production of tumor necrosis factor (Haranaka et al., 1985). Since accumulating evidences indicate that progressive outgrowth of tumor cells is dependent on angiogenesis (Griffioen and Molema, 2000; Holash et al., 1999), these lead us to consider that anti-tumor and anti-metastatic effects of *C. officinale* Makino and *T. bovinus* might be mediated by its angiogenic activities against neovascularization. In the present study, there was a marked reduction of normal branching patterns of blood vessels by treatment of CAM with retinoic acid, *C. officinale* Makino and *T. bovinus*. The retinoic acid-induced inhibition of angiogenesis in CAM is in good agreement with the previous report by Belloni et al. (2000), who reported the inhibitory effects of retinoic acid on the normal outgrowth of capillaries in the CAM. These findings indicate that extracts of *C. officinale* Makino and *T. bovinus* could be active against CAM neovascularization. These anti-angiogenic activities might be one of the pharmacological mechanisms underlying antitumor and antimetastatic properties of *C. officinale* Makino which was suggested by previous reports (Haranaka et al., 1985; Onishi et al., 1998).

Furthermore, to clarify whether *C. officinale* Makino and *T. bovinus* are active against neovascularization in cornea, the rat cornea neovascularization model in which neovascularization was stimulated by suture was used. Oral administration of *C. officinale* Makino and *T. bovinus* markedly reduced the dense vascular networks extending from the corneoscleral limbus and the length of blood vessels in the neovascularized cornea. Taken together, these results suggest that *C. officinale* Makino and *T. bovinus* could be active against neovascular formation in CAM and rat cornea, and anti-tumor and anti-metastatic effects of *C. officinale* Makino and *T. bovinus* may be mediated, in part, by their anti-angiogenic activities.

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